# AI-Based Crop & Agricultural Resource Optimization System

Smart farming through crop recommendation and efficient resource planning

AI + AGRICULTURE

## Choose the right crop.
## Use the right resources.

A decision-support prototype that combines farm conditions, AI-style scoring and resource planning to help reduce unnecessary water and fertilizer use.

SMART FARM

| | |
|---|---|
| Soil | Loamy |
| Water | 500 L/day |
| Rain | 45 mm |
| Temp | 29°C |

# Problem Statement

Why farmers need a simple, data-based planning system

### Crop Choice

Soil, rainfall and temperature can affect which crop is suitable.

### Resource Waste

Water and fertilizer may be wasted without condition-based planning.

### Information Gap

Farm decisions can be difficult when several factors change together.

CORE GOAL

**Reduce resource wastage and support better farm decisions using available farm data.**

# Proposed Solution

From farm inputs to a practical recommendation

### Soil

Soil type & moisture

### Weather

Rainfall & temperature

### Resources

Water availability & area

**AI-BASED ANALYSIS & SCORING**

**Crop Recommendation**

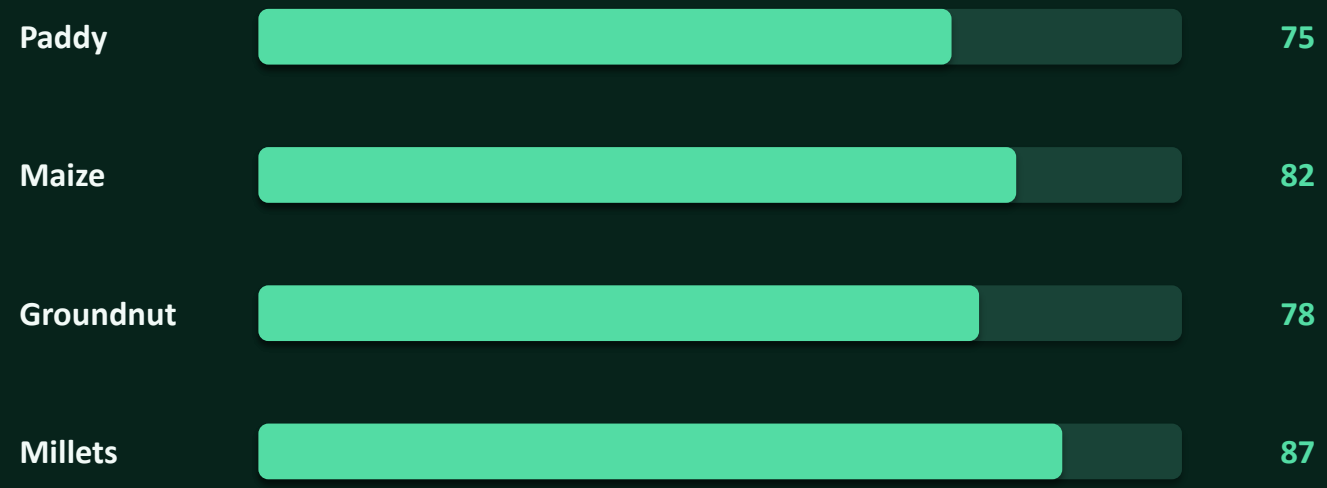
**Resource Plan**

**AI Advice**

# AI Optimization & Innovation

Illustrative suitability scores used by the prototype

## Crop suitability comparison




## What makes it useful?

- ✓ Compares multiple crop options
- ✓ Uses several farm conditions together
- ✓ Connects crop choice with resource planning
- ✓ Produces a simple recommendation

Prototype note: scores are illustrative; real deployment needs validated agricultural datasets and ML models.

# Prototype & Workflow

Interactive HTML dashboard: inputs → analysis → recommendation

1 Farm Data → 2 AI Analysis → 3 Crop Score → 4 Resource Plan → 5 AI Advice

## AI OPTIMIZATION RESULT

Recommended Crop

**MILLET**

Suitability Score

**87 / 100**

Water Plan

**Optimized**

**AI Recommendation**
**Adjust irrigation based on soil moisture and rainfall.**

Prototype: single-page HTML dashboard with interactive inputs and calculated results.

# Impact, Future Scope & Conclusion

Turning farm data into practical resource decisions

### Better Crop Selection

Compare crops using current farm conditions.

### Resource Efficiency

Plan water and fertilizer around available conditions.

### Data-Driven Decisions

Move from fixed assumptions toward measurable inputs.

## FUTURE SCOPE

- Real-time weather APIs
- IoT soil sensors & satellite data
- ML models trained on real datasets
- Regional-language farmer interface

## KEY MESSAGE

**“Use the right crop, at the right time, with the right resources.”**